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PALEOMAGNETISM AND THE ALPINE TECTONICS OF EURASIA III

PALEOMAGNETIC RESEARCH IN THE CENTRAL LEBANON MOUNTAINS AND IN THE TARTOUS AREA (SYRIA)

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SUMMARY

Oriented samples of basalt, basaltic tuff and limestone from the central Lebanon Mountains and the Tartous coastal area were studied with regard to their paleomagnetic characteristics.

The age of the Lebanese samples varies from Upper Jurassic to Lower Cretaceous. The Syrian basalts are from Pliocene age. The initial remanent magnetization vectors were rather scattered, but after cleaning with a.c. fields up to 3,000 Oe they concentrated on the following average directions:

Upper Jurassic : $D = 93°$, $I = 10°$;
Lower Cretaceous: $D = 314°$, $I = 9°$;
Pliocene : $D = 170°$, $I = -35°$.

The Jurassic and the Pliocene direction are considered to be reversed.

These results show a great discrepancy with pole positions already known for Eurasia and for Africa. The deviations may be explained in terms of local tectonic movements or continental drift. The local tectonic disturbances, however, are small.

INTRODUCTION

This study forms part of a general paleomagnetic investigation of the Arabian shield and the Tethys-belt, being carried out jointly by the Utrecht University and the American University of Beirut (Raven, 1964; Gregor and Zijderveld, 1964).

In March 1965 52 oriented samples were taken from nine different locations (fifteen sites) in Lebanon (Fig.1). These sampling sites were widely spread and selected in order to minimize the risk of incorporating errors due to local disturbances. In Syria eight samples were taken from two sites near Tartous from two different lava flows. The names of the locations are given on Table I and on the geological map (Fig.2).

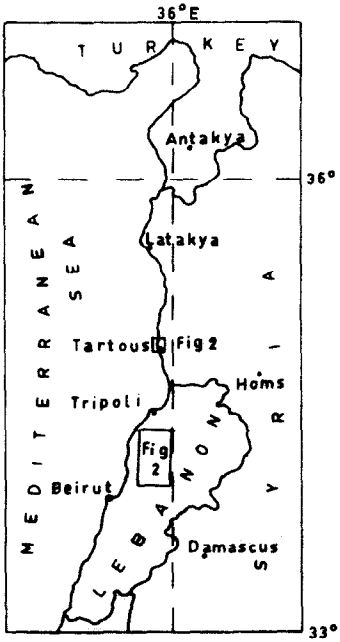


Fig.1. Regional map of the eastern Mediterranean area. Sample areas (Fig.2) are outlined.

GEOLOGICAL SITUATION

The geology of Lebanon and Syria has been studied by several geologists, in particular Dubertret (cf. Dubertret and Wetzel, 1955). The Lebanon, situated at the northern extremity of the African Rift Valley system, presents abundant outcrops of relatively undisturbed Mesozoic rocks, mainly limestones of neritic to deep marine facies. Intermittent volcanism between Late Kimmeridgian and Albian time, concentrated in the central region studied here, has produced basalt flows and tuffs more or less conformably intercalated with the sediments. These features make the region particularly suitable for paleomagnetic research. The Lebanese basalts sampled are of two different ages: Kimmeridgian (j_6) and Aptian (c_2). In addition eight samples were collected from a basaltic tuff in the basal series (c_1) of the Cretaceous (Fig.3 and 4). The sampled limestones from Tourzaya (LIB 22, 23 and 24) are situated above the Aptian basalts. Their age is considered to be still Aptian. The Syrian basalts were dated by Dubertret as Pliocene because he found them intercalated with marine Plaisancian clays (Fig.5).

Fig.2. Geological map of the Lebanese sample area. (After Dubertret and Wetzel, 1955.)

Sample locations:

LIB 1– 7 = Qartaba	LIB 22–24 = Tourzaya	LIB 40–44 = Kfardabian
LIB 10–17 = Laklouk	LIB 25–35 = Mairouba	LIB 47–53 = Hadeth
LIB 19–21 = Ehmej	LIB 37–39 = Reyfoun	LIB 54–56 = Hasroun

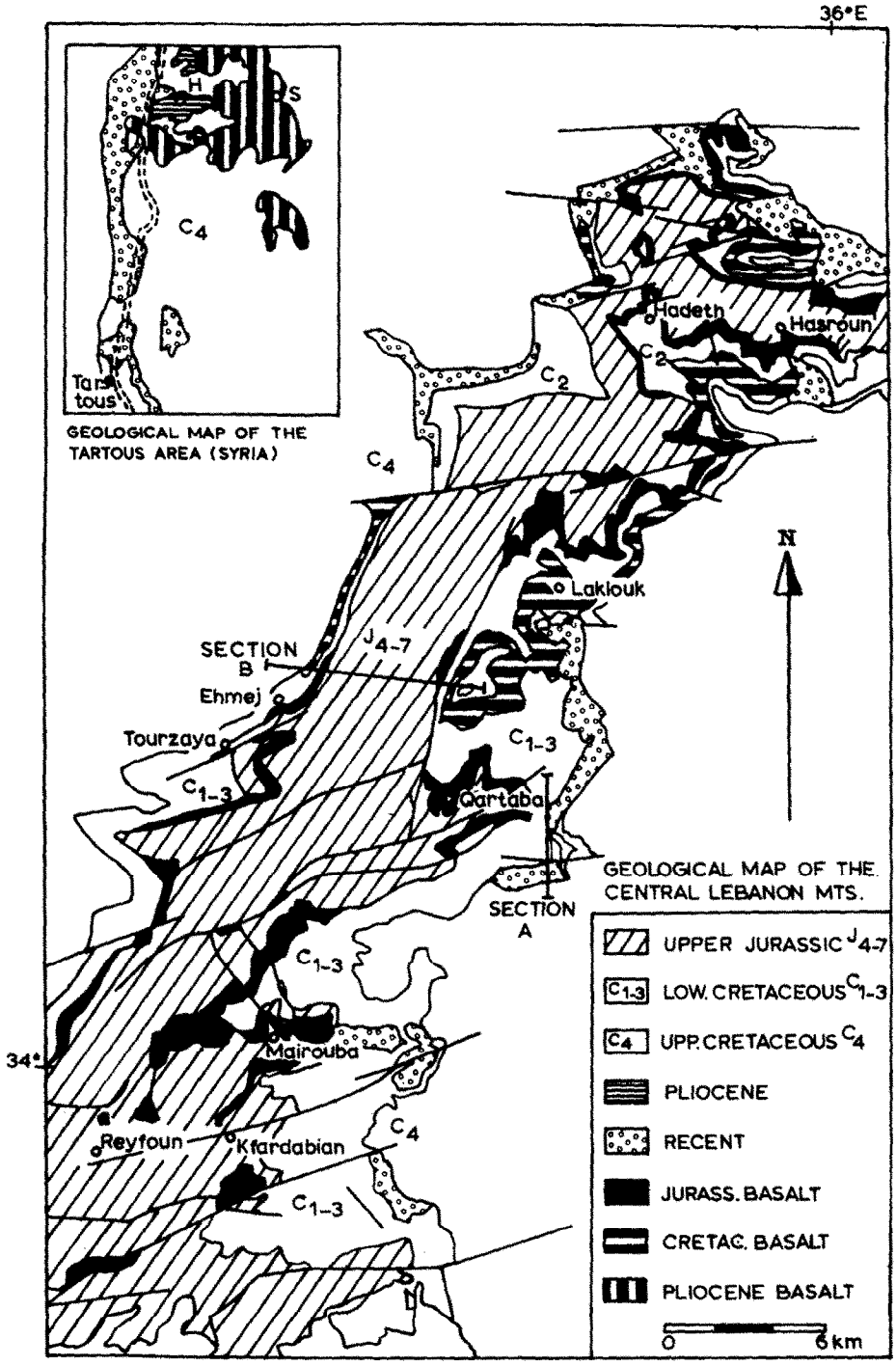

36°E
34°
H
S
C4
Tartous
GEOLOGICAL MAP OF THE TARTOUS AREA (SYRIA)
Hadeth
Hasroun
C2
C4
N
Laklouk
SECTION B
J4-7
Ehmej
Tourzaya
C1-3
Qartaba
SECTION A
GEOLOGICAL MAP OF THE CENTRAL LEBANON MTS.
UPPER JURASSIC J4-7
LOW. CRETACEOUS C1-3
UPP. CRETACEOUS C4
PLIOCENE
RECENT
JURASS. BASALT
CRETAC. BASALT
PLIOCENE BASALT
0
6 km
Mairouba
Reyfoun
Kfardabian

TABLE I

Summary of measurements

Location	Type of rock	Sample no.	Natural remanent magnetization[2]				a.c.-field range
			D[1] (degrees)	I[1] (degrees)	intensity (e.m.u./cm^3 $\times 10^{-6}$)	Q-value[3]	
Kimmeridgian							
I. Qartaba	tuff	LIB 1	95.5	4	860	8.7	300–700
	tuff	LIB 2	86.5	22.5	1,000	5.6	300–700
VI. Ehmej	tuff	LIB 19*	107	−30	450	24.0	
	tuff	LIB 20	76.5	−10	480	2.8	250–500
	tuff	LIB 21*	105.5	−41	240	10.0	
VIII. Mairouba	basalt	LIB 25	78.5	29.5	115	5.1	300–700
	basalt	LIB 26	69	26.5	1,100	5.7	500–1,000
	basalt	LIB 27	85	8.5	2,700	16.0	300–1,000
	basalt	LIB 28	83	23.5	1,150	5.1	300–700
	basalt	LIB 29	78	29.5	520	3.5	700–
	basalt	LIB 30	69.5	35	520	3.5	700–
XI. Reyfoun	basalt	LIB 37	86	23	1,950	8.4	200–500
	basalt	LIB 38	67	34	1,550	6.2	300–500
	basalt	LIB 39	64.5	47	1,630	4.6	300–1,000
XII. Kfardabian	basalt	LIB 40	103.5	42.5	180	2.3	300–700
	basalt	LIB 41	100	22	1,600	3.6	300–750
	basalt	LIB 42	94	13	3,400	8.6	100–300
	basalt	LIB 43	98.5	14	1,850	11.6	100–900
	basalt	LIB 44	105.5	14	450	4.4	100–700
XIII. Kfardabian	tuff	LIB 45*	11.5	53	55	0.7	
	tuff	LIB 46*	65.5	30.5	430	1.7	400–900
XIV. Hadeth ej Joubbe	basalt	LIB 47*	35	59	920	0.8	
	basalt	LIB 48*	22.5	51.5	610	0.6	
XVII. Hasroun	basalt	LIB 54	77.5	19	4,400	2.9	200–500
	basalt	LIB 55	82.5	24	4,400	3.5	100–300
	basalt	LIB 56	78.5	27	3,600	3.0	300–500
Cretaceous - basal series							
XII. Qartaba	tuff	LIB 3*	337.5	51.5	75	1.7	2,800–
	tuff	LIB 4*	18.5	53	120	1.2	2,000–2,900
	tuff	LIB 5*	3	52	105	1.4	2,000–2,900
	tuff	LIB 6*	352.5	55	100	1.4	2,000–2,900
	tuff	LIB 7*	346.5	39.5	90	2.1	1,000–2,900
X. Mairouba	tuff	LIB 33*	346	52.5	50	0.8	2,900–
	tuff	LIB 34*	2.5	49	100	1.3	2,250–
	tuff	LIB 35	350	55	80	0.8	2,800–
Aptian							
IV. Laklouk	basalt	LIB 10	347	43.5	1,650	4.2	400–700
	basalt	LIB 11	358	53	1,700	1.3	100–300
	basalt	LIB 12	330.5	34.5	930	3.0	100–300
	basalt	LIB 13	350.5	45	1,750	2.6	300–700
	basalt	LIB 14	324.5	26	2,300	6.3	200–900

TABLE I (continued)

Components remaining after demagnetization				Mean direction and α_{95} for a given site		
before tectonic correction		after tectonic correction				
D^1 (degrees)	I^1 (degrees)	D^1 (degrees)	I^1 (degrees)	D^1 (degrees)	I^1 (degrees)	α_{95}
94	0.5	94.5	4.5	91.5	8	
90.5	18	88.5	11.5			
87	−19	88.5	5			
101.5	12	101.5	12	94	9	5.5
90.	8	90	8			
89.5	4.5	89.5	4.5			
101	5	101	5			
93.	13	93	13			
88.5	10.5	88.5	10.5			
90.5	13.5	90.5	13.5	88	17	8.75
83.5	15.5	83.5	15.5			
89.5	22	89.5	22			
87.5	28	86.5	20	95.5	13	8.5
100	12	99	4			
93	13	91	12.5			
98.5	14	96.5	15			
106	13.5	104.5	14			
55.5	22	58.5	35.5			
90	8.5	90	8.5	92.5	8.5	3.5
94.5	9	94.5	9			
92.5	8	92.5	8			
319	34.5	311	46			
317.5	32	310.5	44			
343	45.5	340	60			
332.5	49	324	62			
331	29.5	327.5	43			
319.5	43.5	329	27.5			
345	42.5	348.5	23.5			
321	18	323	3.5			
315.5	13	317.5	3	315.5	11.5	8.5
314	31.5	318.5	21			
311	14	313.5	5			
309	12	310.5	3			
317.5	12	318.5	1			

TABLE I (continued)

Location	Type of rock	Sample no.	Natural remanent magnetization[2]				a.c.-field range
			D[1] (degrees)	I[1] (degrees)	intensity (e.m.u. $cm^3 \times 10^{-6}$)	Q-value[3]	
V. Laklouk	basalt	LIB 15	337	32.5	1,600	3.5	300–700
	basalt	LIB 16	305.5	16.5	5,300	35.4	100–900
	basalt	LIB 17	335.5	29.5	1,200	3.3	300–700
VII. Tourzaya	limestone	LIB 22	314	35	2	1.8	2,000–2,700
	limestone	LIB 23	316	27	2	3.6	2,000–2,900
	limestone	LIB 24	314.5	59	2	3.1	2,750–2,900
XV. Hadeth ej Joubbe	tuff	LIB 50	306	13	150	5.7	2,650–
	tuff	LIB 51	303	13	150	5.7	1,600–2,900
	basalt	LIB 53	354.5	44	800	1.1	200–300
Pliocene							
XVIII. Saouda	basalt	SYR 1	41.5	−43.5	240	2.8	200–1,500
	basalt	SYR 2	16	−24	1,050	8.9	200–700
	basalt	SYR 3	123	−50	770	8.6	100–250
	basalt	SYR 4	20.5	−41.5	1,350	14.3	200–1,000
XIX. Housseine	basalt	SYR 5	197.5	−43	4,700	19.0	100–600
	basalt	SYR 6	195.5	−39.5	4,650	23.0	50–2,000
	basalt	SYR 7	167.5	−29.5	2,250	16.0	100–1,500
	basalt	SYR 8*	65.5	28.5	2,550	14.1	100–2,000

[1]D = declination; I = inclination.
[2]Corrected for present local magnetic declination (March 1965: 2.5°)
[3]Q-value = $J/k \cdot H_Z$, in which J = intensity in e.m.u./cm^3, k = susceptibility and H_Z = vertica component of the ambient field.
*Not used for computation.

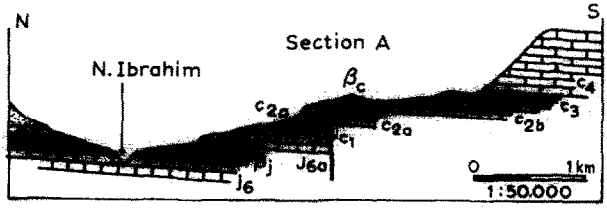


Fig.3. Geological section, east of Qartaba, showing the stratigraphic sequence of Upper Jurassic and Lower Cretaceous with basaltic intercalations (black). (After Dubertret and Wetzel, 1955.)

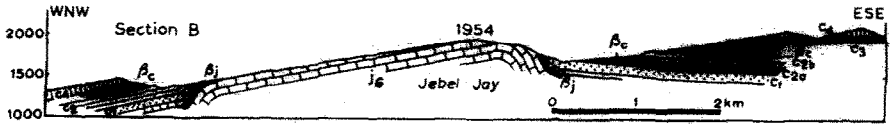


Fig.4. Geological section southwest of Laklouk. (After Dubertret and Wetzel, 1955.)

TABLE I (continued)

Components remaining after demagnetization				Mean direction and α_{95} for a given site		
before tectonic correction		after tectonic correction				
D^1 (degrees)	I^1 (degrees)	D^1 (degrees)	I^1 (degrees)	D^1 (degrees)	I^1 (degrees)	α_{95}
321.5	17.5	321.5	17.5	315	16.5	14
304.5	14.5	304.5	14.5			
319.5	17	319.5	17			
313	24.5	312	11.5	315	17	19
314.5	22	313	9.5			
324.5	42.5	321	30.5			
301	11	301.5	10			
306.5	6	307	6			
307.5	2.5	307.5	−10			
171.5	−50	171.5	−50	170	−32.5	15
170.5	−25	170.5	−25			
162.5	−24.5	162.5	−24.5			
177	−30.5	177	−30.5			
201.5	−41	182	−45	170	−41	24
199	−44.5	178	−47			
165.5	−37	153.5	−29.5			
53	20	45	35.5			

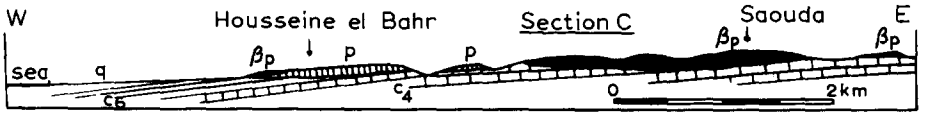
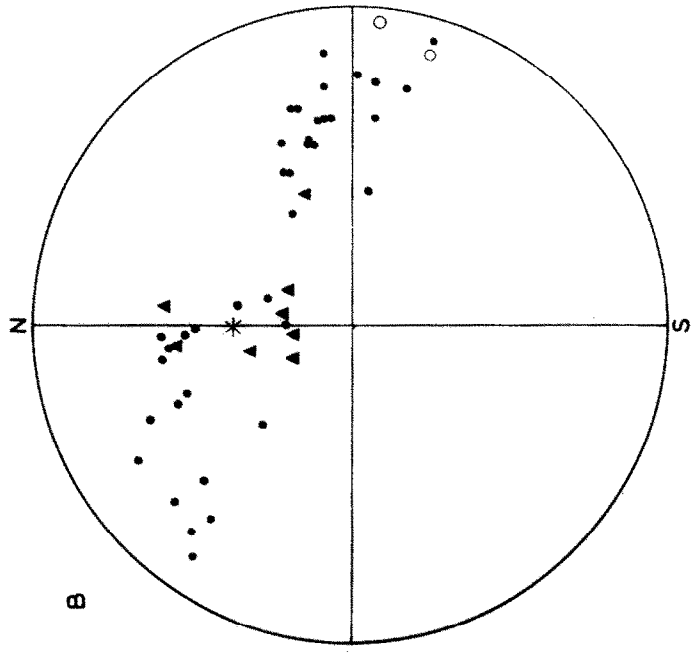


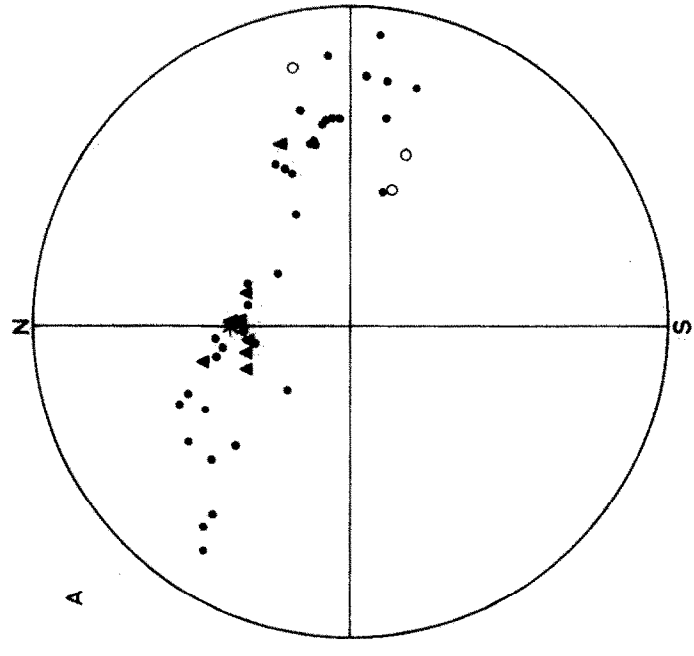
Fig.5. Geological section north of Tartous, Syria. (After Dubertret and Wetzel, 1955.)

q = Quaternary; p = Pliocene; $c_{4,6}$ = Upper Cretaceous, β_p = Pliocene basalts (black).

METHODS OF RESEARCH

The samples were sawn to approximately cubic shape in order to minimize shape-effects and embedded in their correct orientation in cubes of paraffin with 10 cm edges; directions and intensities of magnetization were then measured on the Utrecht astatic magnetometers. For an extensive description of the methods of measurement and demagnetization we refer to As and Zijderveld (1958) and Van Everdingen (1960). The data thus obtained are summarized in Table I and plotted in stereograms (Fig.6 and 9).


N
S
B


N
S
A

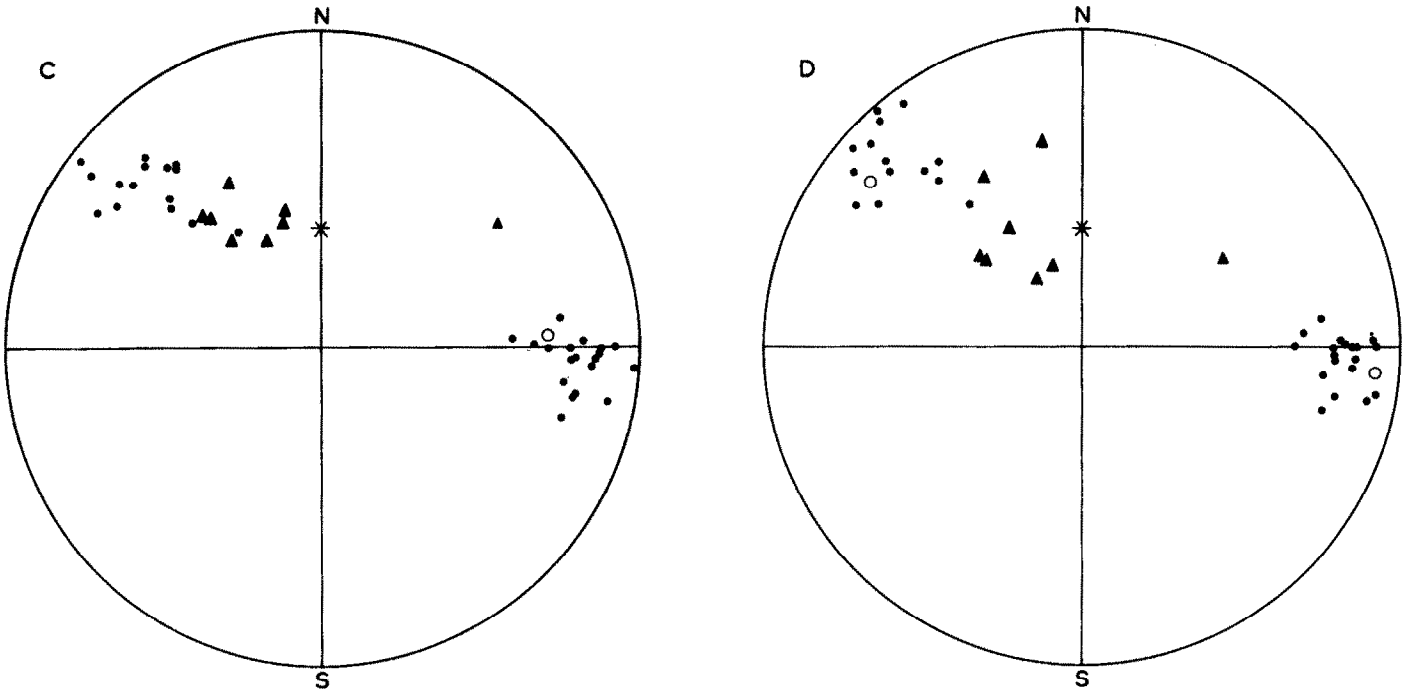
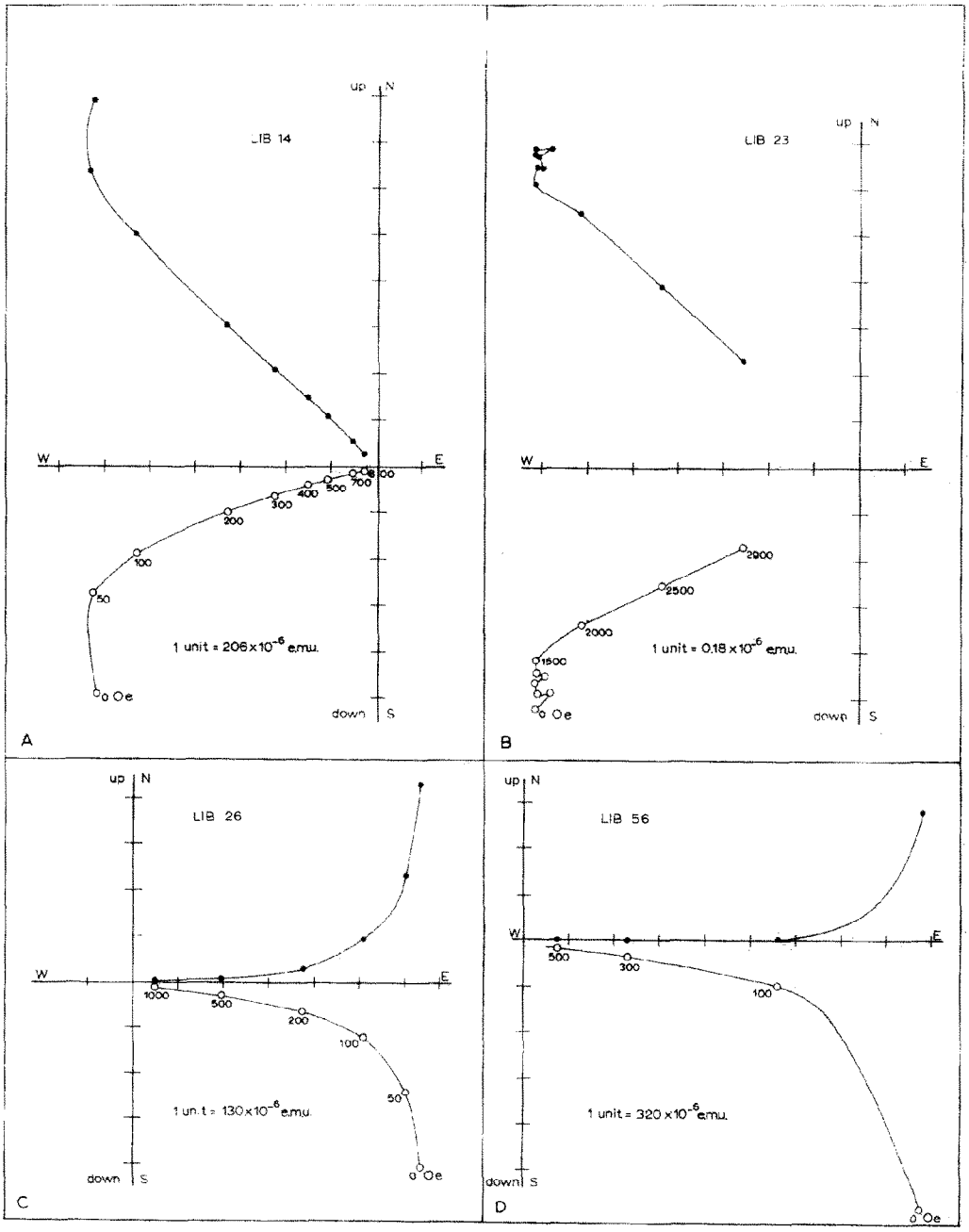


Fig.6. Stereograms showing the directions of the magnetic vectors of the Lebanese samples.
A. Initially measured natural remanent magnetization (N.R.M.), without tectonic correction.
B. Initially measured N.R.M., after tectonic correction.
C. Components after magnetic cleaning with a.c. magnetic fields up to 3,000 Oe, without tectonic correction.
D. Components after magnetic cleaning with a.c. magnetic fields up to 3,000 Oe, after tectonic correction.
O = directions pointing upward (north-seeking); ● = directions pointing downward (north-seeking); ▲ = tuffaceous samples, unreliable due to weathering (mostly with a strong secondary component which conforms to the present-day local magnetic field); × = local direction of the axial earth-magnetic dipole field.


LIB 14
up
N
W
E
S
down
700
500
400
300
200
100
50
0 Oe
1 unit = 206 x 10⁻⁶ e.m.u.
A
LIB 23
2500
2000
1500
1 unit = 0.18 x 10⁻⁶ e.m.u.
B
LIB 26
1000
500
200
100
50
1 unit = 130 x 10⁻⁶ e.m.u.
C
LIB 56
500
300
100
1 unit = 320 x 10⁻⁶ e.m.u.
D

RESULTS OF THE LEBANESE SAMPLES

As shown in Fig.6A, the natural remanent magnetizations (N.R.M.) are rather scattered, but two groups can be distinguished fairly easily: one (Kimmeridgian) with an easterly declination and another (Lower Cretaceous) with a northwesterly declination.

The samples were progressively demagnetized with the aid of a.c. magnetic fields up to 3,000 Oe. The results of this are illustrated in Fig.7 and 8.

The three types of rocks (basalts, tuffs and limestones) did not react identically to the progressive demagnetization. The basalts lost their unstable component in an a.c. magnetic field of about 200 Oe and showed a total loss of remanent magnetization at 1,000 Oe (Fig.7). They had a very strong N.R.M. (intensity $5–50 \cdot 10^{-4}$ e.m.u./cm^3) and also high Q-ratios (Königsberger, 1938). The secondary component, easily eliminated, is probably induced by the present-day local geomagnetic field (Fig.7). On treatment with a.c. magnetic fields above 200 Oe the decrease of the magnetization vector takes place in a direction that passes straight through the origin. This fact shows that after the elimination of the secondary magnetization, only one component was left. This latter was found to have the same direction for all basalt samples of the same age, with the exception of a few weathered samples in which it did not show up very well, being replaced almost entirely by a strong secondary component presumably as a result of the weathering.

The tuffs gave more difficulty. Because they are more easily weathered, the present-day local geomagnetic field adds a strong secondary component, which completely dominates the primary component. However, when the samples are progressively demagnetized there is still an indication that their characteristic magnetization[1] has about the same direction as that of the contemporaneous basalts. In most cases the tuffs had not completely lost their remanent magnetism after treatment with a.c. magnetic fields up to peak value 3,000 Oe. For these reasons we have discarded some tuffaceous samples as being unreliable and did not use them for calculating the average pole position (LIB 3, 4, 5, 6, 7, 33, 34; in Fig.6 they are indicated by triangles. Microscopic examination of thin slices of these samples revealed a strong alteration, compared with the other tuffs). The intensity of the N.R.M. of the basaltic tuffs varied between $1 \cdot 10^{-4}$ and $10 \cdot 10^{-4}$ e.m.u./cm^3.

[1]Where conclusive proof of the primary character of the residual component revealed by progressive demagnetization is lacking, we adopt the term "characteristic magnetization" for this component. When we find the same residual component in several samples from a given site, we speak of the characteristic magnetization of that site. The same concept can be extended quite naturally to a regional scale (Gregor and Zijderveld, 1964).

Fig.7. Demagnetization diagrams. A. Aptian basalt. B. Aptian limestone (showing the advantage of demagnetizing up to 3,000 Oe). C and D. Kimmeridgian basalts.

Plotted points represent successive positions - in orthogonal projection - of the end of the magnetic vector during progressive demagnetization. ● = projection on the horizontal plane; o = projection on the east–west vertical plane.

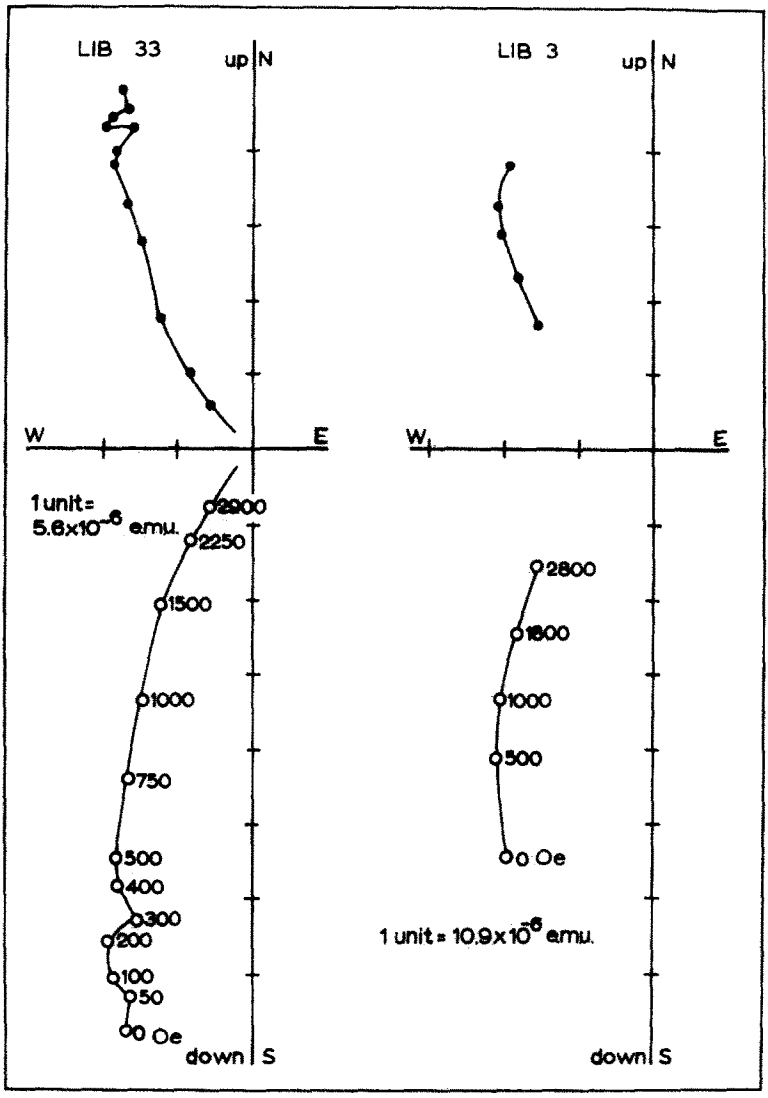


Fig.8. Demagnetization diagrams. Samples of Lower Cretaceous tuffs, showing a strong secondary magnetization conform the present-day local earth-magnetic field, due to weathering. Owing to the unreliability of the characteristic component, the samples LIB 3, 4, 5, 6, 7, 33 and 34 are excluded from the calculation of the average pole position.

Plotted points represent successive positions - in orthogonal projection - of the end of the magnetic vector during progressive demagnetization. ● = projection on the horizontal plane; o = projection on the east–west vertical plane.

The N.R.M. of the pink limestones of Aptian age (stratigrafically above the Aptian basalts) consists of a very hard characteristic magnetization and a small secondary component which is eliminated in a.c. magnetic fields below 1,500 Oe. The intensity of the total N.R.M. of these limestones was $2 \cdot 10^{-6}$ e.m.u./cm^3. The direction of their characteristic magnetizations was entirely consistent with the direction of the Aptian basalts.

RESULTS OF THE SYRIAN SAMPLES

The directions of the N.R.M.'s are plotted in Fig.9A. With the exception of SYR 8 all samples reveal a more or less similar characteristic magnetization in the process of demagnetization (Fig.9).

The samples from the location near Saouda (SYR 1–4) yielded a rather remarkable result. At least three magnetic directions were found in the progressive demagnetization. A very soft component is already removed below an a.c. magnetic field of 100 Oe. A second is found between values of 100 and 1,000 Oe and a third between 1,000 and 3,000 Oe. This third we regard as a very hard secondary component as we found its direction conforms with the present-day local geomagnetic field (Fig.10). The second component is very consistent with the characteristic component of the samples SYR 5–7 and therefore also considered to be the characteristic magnetization (Fig.9C). The aberrant direction (SYR 8) is probably due to a mistake while sampling and therefore not included in the calculation of the average direction.

CONCLUSIONS

Considering the results summarized in the last four columns of Table I, we see that from all 20 samples of Kimmeridgian age the characteristic magnetization concentrates on an easterly direction with a small positive inclination. Fourteen samples of the Aptian and one sample of the Cretaceous basal series have a northwesterly direction with a small positive inclination. These results are plotted in Fig.6D in stereographic projection.

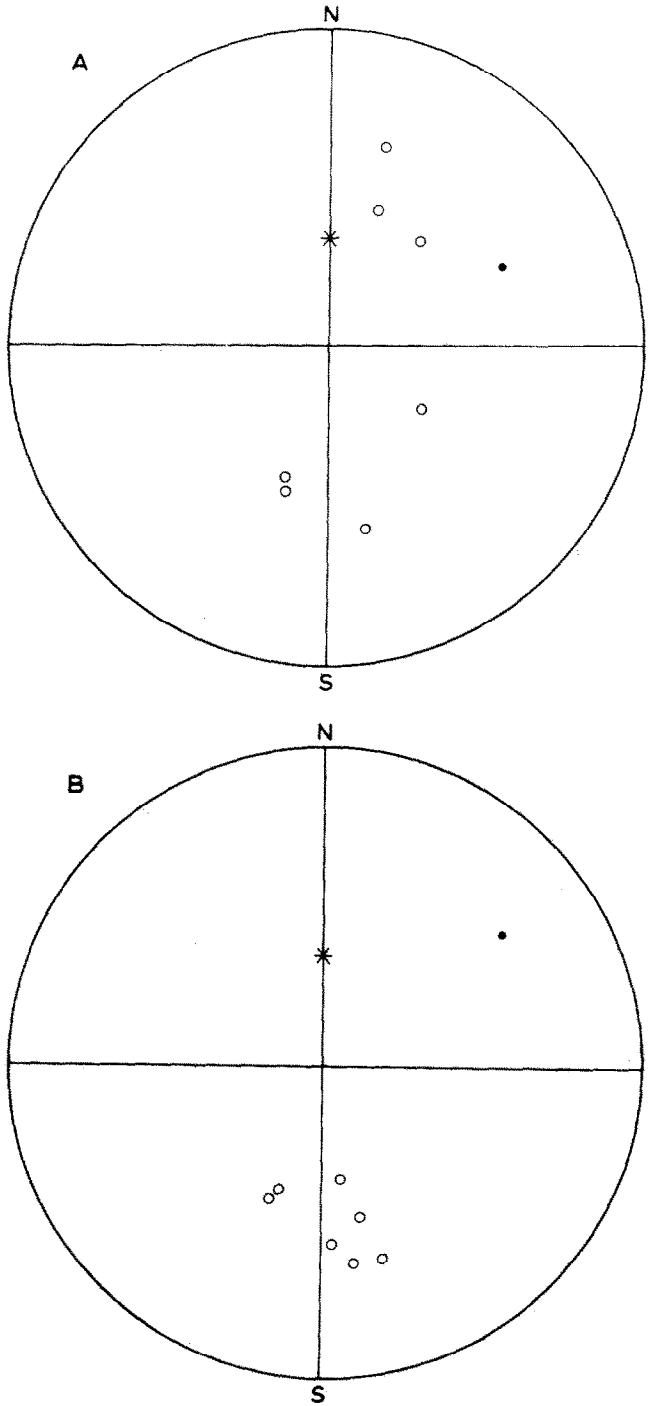
In Table II the average directions of magnetization are given and for

TABLE II

Mean directions of magnetization and ancient pole positions[1]

Period	Location	N_1	N_2	D (degrees)	I (degrees)	α_{95} (degrees)	Ancient pole position
Kimmeridgian	Lebanon	6	20	92.5	10.5	3	120 °E 1°N
Aptian	Lebanon	5	15	313.5	9.5	5.5	78 °W 38°N
Pliocene	Syria	2	7	170	−35.5	10	71 °E 73°S
Lower Jurassic	Africa[2]	74					81.5°W 67°N
Upper Cretaceous	Africa[3]	7					101 °W 62°N

[1] N_1 = number of sites, N_2 = number of samples, D = declination after demagnetization and tectonic correction, I = inclination after demagnetization and tectonic correction, α_{95} = semi-angle of cone of 95% confidence for mean direction.
[2] After Van Zijl et al. (1962).
[3] After Gough and Opdyke (1963).


A
N
S
B
N
S

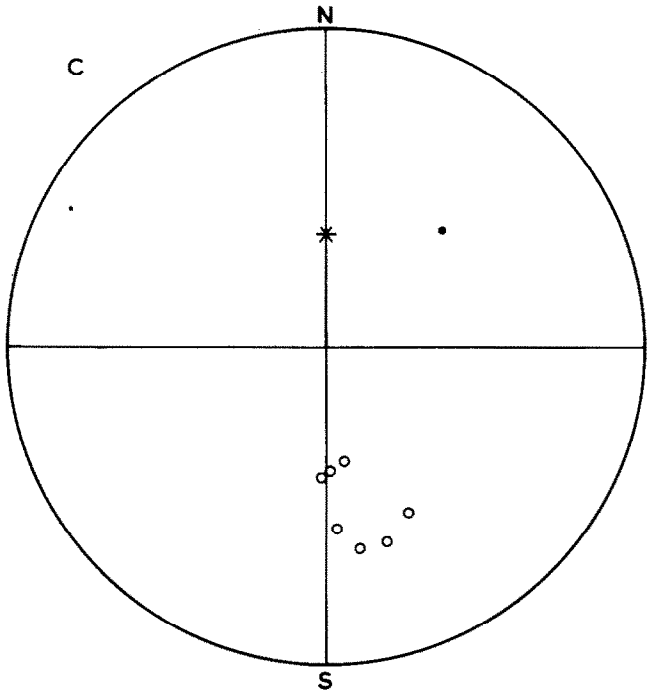


Fig.9. Stereograms showing the directions of the magnetic vectors of the Pliocene basalts of Syria, north of Tartous.

A. Initially measured natural remanent magnetization (N.R.M.), without tectonic correction.

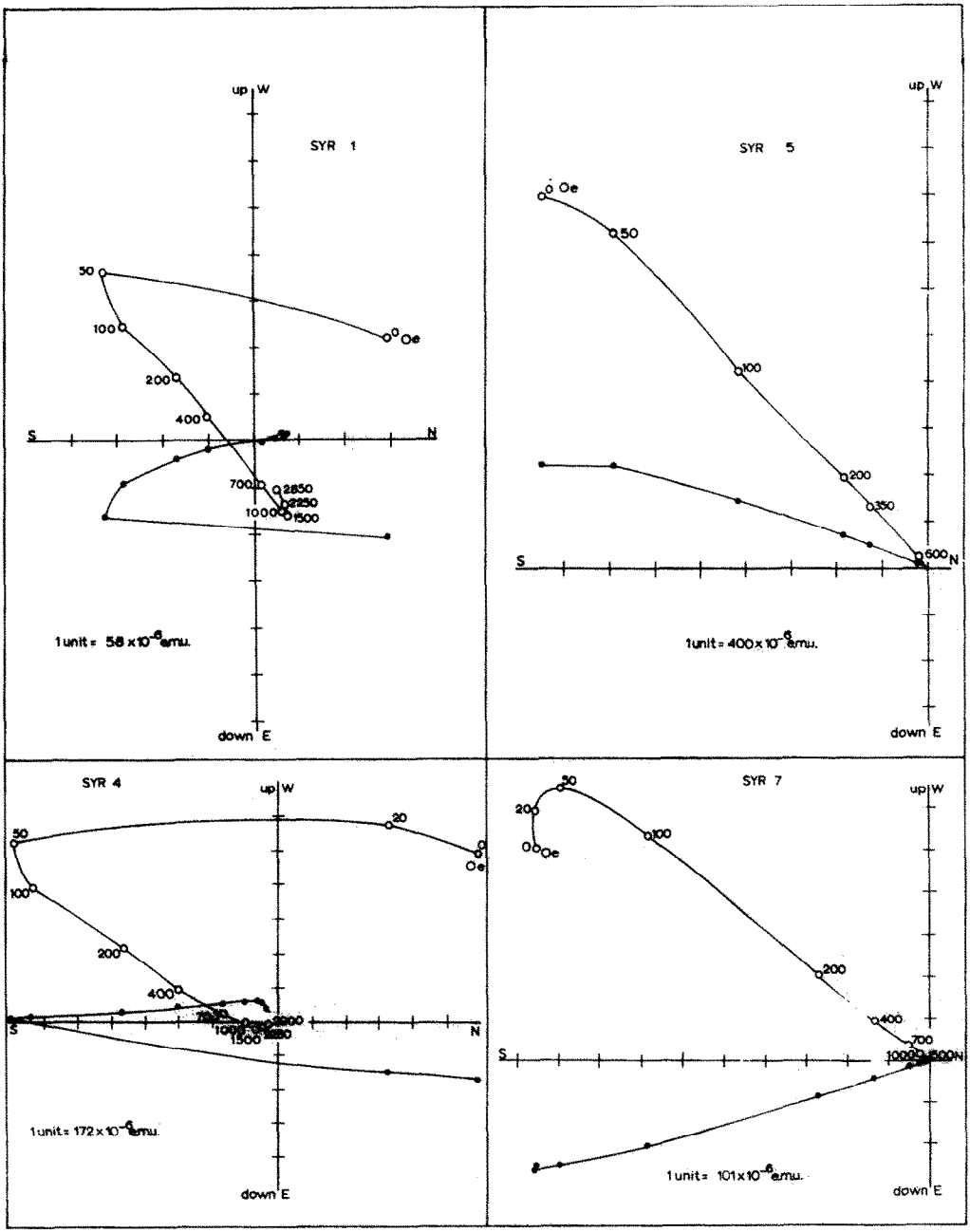
B. Components after magnetic cleaning with a.c. magnetic fields up to 3,000 Oe, without tectonic correction.

C. Components after magnetic cleaning with a.c. magnetic fields up to 3,000 Oe, after tectonic correction.

o = directions pointing upward (north-seeking); ● = directions pointing downward (north-seeking); ⁎ = local direction of the axial earth-magnetic dipole field.

comparison some reliable data from Africa are included (see also Fig.11). In seeking a reasonable interpretation of our data, we must consider: (1) reversals of the earth's magnetic field; (2) secular variation; (3) local tectonics; (4) the (regional) tectonic relation of the Lebanon to Africa; (5) continental drift; (6) polar wandering.

(Ad.1) In order to circumvent the implication of a shift of the pole position over some 140°, accomplished in the time-span Kimmeridgian–Aptian of at most 15 million years, this being evidently preposterous, we regard our Kimmeridgian direction as reversed, i.e., we consider 120°E 1°N to be the south pole. This first assumption is not conflicting with known data from the Lower Jurassic (e.g., Gough et al., 1964) as well as from the Upper Jurassic (Irving, 1964). As a first approximation, this assumption allows us to plot the


SYR 1
up W
50
100
200
400
0 Oe
S
N
700
1000
2850
2250
1500
1 unit = 58 x 10⁻⁶ e.m.u.
down E
SYR 5
up W
0 Oe
50
100
200
350
600
S
N
1 unit = 400 x 10⁻⁶ e.m.u.
down E
SYR 4
up W
50
100
200
400
20
0 Oe
S
N
1500
1 unit = 172 x 10⁻⁶ e.m.u.
down E
SYR 7
up W
50
20
0 Oe
100
200
400
700
1000
S
N
1 unit = 101 x 10⁻⁶ e.m.u.
down E

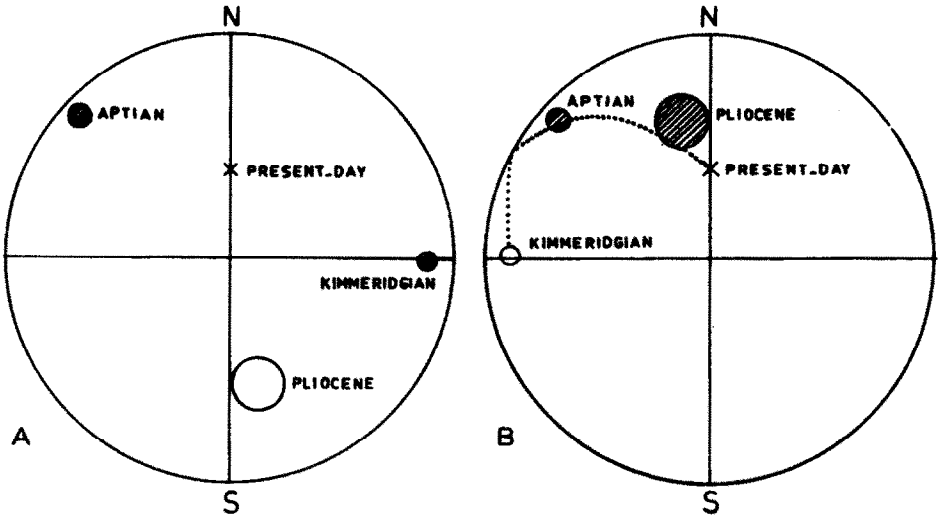


Fig.11. Schematic stereograms. A. Computed geomagnetic directions with circles of confidence for Kimmeridgian, Aptian and Pliocene (see Table II). B. The same, with reversed directions of Kimmeridgian and Pliocene.
..... = path of the changing geomagnetic direction in Lebanon and Syria from Kimmeridgian till present-day; o = directions pointing upward; ● = directions pointing downward.

change of direction of the local geomagnetic field as the dotted line in Fig.11B. The N.R.M. direction of the Pliocene basalts from Syria is also reversed.

(Ad.2) Owing to the small number of Pliocene samples investigated we cannot attach much significance to our Pliocene pole. It will be seen, however, that in reconstructing the area's local structural disturbance or drift, our result does not contradict earlier paleomagnetic data (Irving and Tarling, 1961). When we consider the average Aptian pole position, calculated from fifteen samples (from five sites of various basaltic flows and one site of limestone) we may exclude the possibility that all our fifteen samples acquired their primary magnetization within a geologically short period and may thus exclude the treacherous effect of secular variation.

Fig.10. Demagnetization diagrams of Pliocene basalts, north of Tartous, Syria. Samples SYR 1 and SYR 4 show a weak unstable component, which is removed in a.c. magnetic fields below 50 Oe, a characteristic reversed direction and a "normal" direction, conformable to the present-day local geomagnetic field. Samples SYR 5 and SYR 7 lose their weak component also below 50 Oe, and then show a stable magnetization; the direction of their decrease passes straight through the origin.

Plotted points represent successive positions - in orthogonal projection - of the end of the magnetic vector during progressive demagnetization. ● = projection to the horizontal plane; o = projection on the north–south vertical plane.

Unfortunately Kimmeridgian limestones are too pure for paleomagnetic investigations and as long as no other data from the immediate neighbourhood are available, the remarkable small scatter of our Kimmeridgian directions must be a warning not to give these samples the same weight as the Aptian ones: all our Kimmeridgian samples might conceivably be from one very short period of volcanism.

(Ad.3 and 4) Previous investigators have established a continuity of facies and a gradually diminishing thickness of Upper Mesozoic sediments from the Lebanese coast far into Syria; we may conclude that our area formed a marginal basin of the Arabian Shield at least after the Triassic period. Moreover, as Lebanon's gentle folds show very little compression, relative movements between our area and the Arabian Shield, due to crustal shortening can not have been more than some tens of kilometers. If we now once more consider the south-pole positions calculated from the Aptian of the Lebanon and the African results (Table II), we may deduce: (a) without any displacement of Lebanon relative to Africa a distance of 28° between the Upper Cretaceous pole position for Africa and our Aptian pole position; (b) a distance of 19° between the Upper Cretaceous pole position for Africa and our Aptian pole position, accepting a 7° anticlockwise rotation suggested by Irving and Tarling (1961) for their data from Aden and a northward translation of the Arabian block relative to Africa (Irving and Tarling, 1961) over a distance of about 150 km; Quennel (1958) deduced 107 km from his geomorphologic and tectonic studies on the Dead Sea rift valley.

Our Kimmeridgian results and the Jurassic data for Africa are completely out of harmony with each other; there exists a distance of 70° between their ancient pole positions.

Points 5 and 6 must be left out of consideration in this paper.

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